REPORT

Semantic Analysis of IPCC AR6 WGII Chapter 4: Water

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in partial fulfilment of the requirements of
IASc-INSA-NASI SRFP 2025

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1. Introduction

Climate change is no longer a distant threat, its impacts are visible across ecosystems, economies, and communities worldwide. Recognizing the urgency, the global scientific community established the Intergovernmental Panel on Climate Change (IPCC) under the United Nations and the World Meteorological Organization to provide comprehensive assessments of climate science, impacts, and response options.

The Sixth Assessment Report (AR6) is the most recent and detailed effort by the IPCC. It consists of contributions from three Working Groups and culminates in the 2023 Synthesis Report, described by UN Secretary-General António Guterres as "the survival guide for humanity." The Synthesis Report distills the findings of the three major reports:

- Working Group I (WGI) – The Physical Science Basis
- Working Group II (WGII) – Impacts, Adaptation, and Vulnerability
- Working Group III (WGIII) – Mitigation of Climate Change

These reports are deeply interconnected, and together they form a foundation for science-based policy action. The Synthesis Report brings their key messages together into a unified narrative, offering a holistic overview of the climate crisis and the urgent steps needed.

Among the various chapters available across the IPCC AR6 reports, I chose to work on Chapter 4 of Working Group II – Water[1].

Summary of IPCC AR6 WGII Chapter 4: Water

This chapter presents a comprehensive assessment of how climate change is affecting the global water cycle. It explores observed and projected changes in precipitation, snow and ice melt, droughts, floods, and water availability, along with their consequences for human societies and ecosystems.

Key highlights from the chapter include:

- **Intensification of the water cycle**: Global warming is leading to more intense rainfall events and longer dry spells, disrupting agricultural practices and water supply systems.
- **Widespread water-related risks**: At 2°C of global warming, between 0.9 to 3.9 billion people are projected to experience increased water stress. Vulnerabilities are especially high in regions with limited adaptive capacity.
- **Impacts on multiple sectors**: Water-related climate risks directly affect agriculture, sanitation, urban planning, public health, energy production, and biodiversity.
- **Adaptation and responses**: Over 60% of documented climate adaptation strategies globally focus on water-related issues, demonstrating the central role of water in climate resilience planning.

By working on this chapter, I aimed to extract meaningful keywords, build a domain-specific dictionary, and create an interactive semantic structure that could help scientists, educators, and the public explore the content more effectively.

2. Aims and Objectives

My work under the #semanticClimate initiative was guided by the broader goal of transforming climate-related documents, especially the IPCC's Sixth Assessment Reports into semantic, structured, and accessible formats. The intention is to make these critical scientific materials easier to explore, search, and understand by both humans and machines. This is essential for increasing engagement, promoting open science, and enabling global action on climate change.

During the internship, my objectives were focused on the following key areas:

- **Understanding the Project and Tools:**

Before contributing meaningfully to the project, it was important to first understand the software ecosystem and the vision behind #semanticClimate. The tools developed by the team are designed to process large scientific documents and convert them into machine-readable, semantically enriched content. I spent time exploring the core libraries, attending team discussions, reviewing documentation, and setting up the development environment. This helped me align my work with the project's values of open knowledge, transparency, and collaborative science.

- **Keyword Extraction:**

Identifying and extracting relevant keywords from climate reports is essential because these terms represent the scientific concepts, phenomena, and concerns discussed in the chapter. They form the basis for glossaries, semantic tagging, and information retrieval. My task involved analyzing the chapter's content to generate a curated list of domain-specific keywords. These terms capture the essence of the chapter and support downstream tasks like dictionary building and knowledge graph construction.

- **Dictionary Development:**

A dictionary serves as a bridge between technical content and public understanding. It provides clear, contextual definitions that help readers grasp complex scientific terminology. After generating the keyword list, I created a structured dictionary that linked each term with accessible descriptions. The goal was to enhance comprehension for non-expert audiences and support

multilingual translation and educational use. This dictionary also contributes to semantic interoperability by integrating linked open data.

- **Knowledge Graph Design:**

Climate reports often contain dense and complex structures, making them difficult to navigate. Creating a knowledge graph helps visualize the hierarchy and relationships between sections, enabling intuitive exploration of the document. I constructed a graph that reflected the structure of Chapter 4, highlighting its major themes and subtopics. This serves as a visual map for readers and can also be embedded into interactive platforms for enhanced digital engagement.

- **Alpha testing tools:**

As part of ensuring the reliability and usability of the tools developed under the #semanticClimate initiative, I participated in alpha testing key software components. This involved running the tools on sample chapters, reporting bugs, and suggesting improvements based on user experience. The focus was on validating outputs such as keyword lists, dictionaries, and semantic markup. I contributed feedback to enhance the robustness and accessibility of the tools for future users. Through this process, I gained experience in software quality assurance and helped ensure that the tools meet the standards of accuracy, usability, and open collaboration.

Each of these objectives contributes to #semanticClimate's larger mission of building an open, semantic infrastructure for climate science—empowering users to interact with critical data in meaningful and accessible ways.

3. Activities & Accomplishments

The core of my internship work focused on semantically enriching Chapter 4 (“Water”) of the IPCC AR6 Working Group II report. This involved a series of technical and research-driven tasks aimed at making the chapter's content more accessible, machine-readable, and interoperable. By extracting key scientific terms, creating structured dictionaries, and visualizing the chapter's structure as a knowledge graph, I contributed to building the foundational components necessary for advanced search, retrieval, and educational applications. These processes were aligned with #semanticClimate’s mission to enable open, transparent, and reusable climate knowledge.

3.1 Wordlist Extraction for IPCC AR6 WG2 Chapter 4

The first step in enriching the content semantically was the extraction of domain-specific keywords from Chapter 4 ("Water") of the IPCC AR6 WGII report. This chapter discusses critical water-related climate issues, and identifying key terms within it is essential for tasks such as tagging, glossary generation, and knowledge mapping.

- **Downloaded HTML File:** Acquired the cleaned and preprocessed HTML version of Chapter 4 from the IPCC content repository. This version was specifically formatted to support machine reading and downstream processing.
- **Environment Setup:** Created a virtual environment using Anaconda Navigator to ensure isolation and compatibility. Installed necessary Python dependencies.
- **Keyword Extraction:** Utilized a custom script (keyword_extraction.py) to preprocess and analyze the chapter text. This included tokenization, stopword removal, and frequency analysis. The output was a curated wordlist capturing important scientific and technical terms related to water and climate.
- `python keyword_extraction.py -t text_file_path.txt -s ./`

196 lines (195 loc) · 3.21 KB

Code | Blame

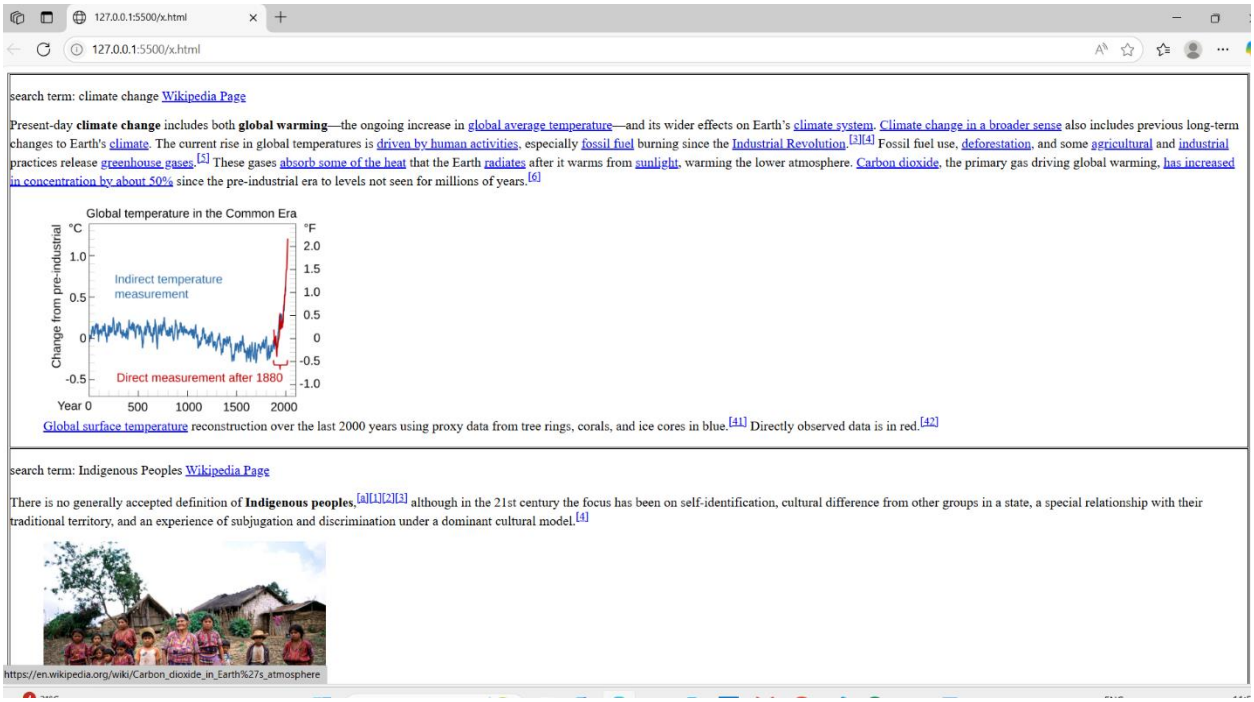
```
climate change
Indigenous Peoples
precipitation
anthropogenic climate change
hydropower production
sanitation
permafrost
hydropower
migration
Mediterranean
streamflow
agronomic practices
precipitation
Desalination
river basins
Anthropogenic climate change
desertification
sediment load
coping strategies
Intergovernmental Panel
wetlands
maladaptive outcome
Geoforum
aquifers
vulnerable communities
Evapotranspiration
terrestrial ecosystems
```

Extracted wordlist

3.2 Dictionary Creation for IPCC AR6 WG2 Chapter 4

The second major task was transforming the extracted keywords into a structured semantic dictionary. This dictionary aims to bridge the gap between complex scientific language and public comprehension by linking each term to an accessible definition.

- **Input Preparation:** Processed the generated keyword list and ensured the terms were contextually relevant and distinct.
- **Dictionary Generation:** Used the amilib framework to link each keyword to its corresponding description sourced from Wikipedia. The dictionary was produced in HTML format for easy integration into browsers and educational platforms.
- **Optional Features:** Included semantic figures where available.
- amilib DICT --words your_wordlist_path.txt wordlist.txt --description wikipedia --dict output_dict_path.html --figures --operation create

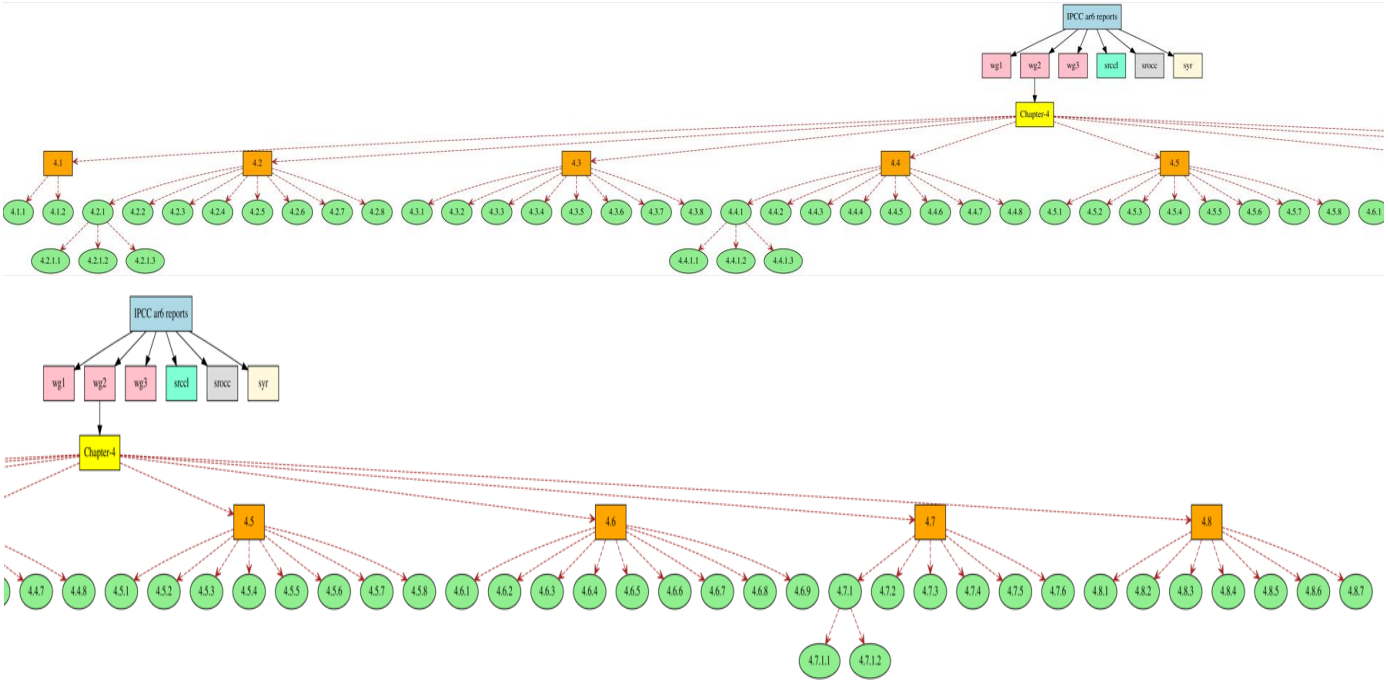


Dictionary Created

3.3 Knowledge Graph Creation

This graph presents a hierarchical and thematic overview of the chapter's contents, helping both technical and non-technical users navigate complex material intuitively.

- **Table of Contents Extraction:** Parsed the section headers and subheaders from the chapter to understand the internal structure and thematic flow.
- **Graph Design:** Designed a semantic knowledge graph using the Graphviz library. Each node represents a section or subsection, while the edges indicate logical or thematic relationships. Dotted red edges were used to signify implicit or loosely connected topics. Tooltips were added to provide additional context when hovering over nodes.
- **Output Generation:** The final graph was rendered as an SVG (ipcc_ch04_graph.svg) for embedding into Jupyter Notebooks, technical documents, or web interfaces.



Knowledge Graph created using Graphviz

3.4 Alpha testing of tool

Contributed to improving tool robustness by participating in early-stage testing (alpha phase) of key Semantic Climate tools.

- **Tested Tools:**
  - Pygetpapers (for scientific paper retrieval)
  - amilib (for semantic enrichment and dictionary creation)
  - LLMRAG (for AI-assisted document retrieval and analysis)
- Reported bugs, usability issues, and provided enhancement suggestions during internal testing phases.
- Verified outputs against expected results and helped validate tool behavior under different input scenarios.

```
Ran 7 tests in 612.267s

OK

(venv) C:\Users\Dell\llmrag>
(venv) C:\Users\Dell\llmrag> coverage report -m
Name                                                    Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------------
llmrag\chunking\__init__.py                                 1      0   100%
llmrag\chunking\text_splitter.py                            8      0   100%
llmrag\cli.py                                              52     52     0%   1-81
llmrag\embeddings\__init__.py                               3      0   100%
llmrag\embeddings\base_embedder.py                          5      1    80%   6
llmrag\embeddings\sentence_transformers_embedder.py        12      0   100%
llmrag\generators\__init__.py                               0      0   100%
llmrag\generators\local_generator.py                       11      7    36%   16, 19-35
llmrag\main.py                                             56     56     0%   1-72
llmrag\models\__init__.py                                   3      0   100%
llmrag\models\base_model.py                                 5      1    80%   6
llmrag\models\transformers_model.py                         8      0   100%
llmrag\pipelines\__init__.py                                1      0   100%
llmrag\pipelines\rag_pipeline.py                           27      6    78%   19, 21, 61-64
llmrag\retrievers\__init__.py                              12      7    42%   7-14
llmrag\retrievers\base_vector_store.py                      8      2    75%   6, 10
llmrag\retrievers\chroma_store.py                          42      3    93%   48, 67-68
llmrag\retrievers\faiss_store.py                           27     27     0%   1-31
llmrag\streamlit_app.py                                    29     29     0%   1-53
-------------------------------------------------------------------------------------
TOTAL                                                     310    191    38%
```

output obtained when run unittest on llmrag

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================================================================== short test
 summary info ============================================================================
===
FAILED test/test_extract_text.py::ExtractTextTest::test_keybert_breward_1 -
IndexError: list index out of range
FAILED test/test_file.py::File0Test::test_list_children - AssertionError: as
sert ['C:\\Users\\...files\\file1'] == ['/Users/pm28..._files/file1']
FAILED test/test_file.py::File0Test::test_relative_pathname - AssertionError
: assert 'd\\e.txt' == 'd/e.txt'
FAILED test/test_html.py::CSSStyleTest::test_tinycss - ModuleNotFoundError:
No module named 'tinycss'
FAILED test/test_misc.py::ArgsTest::test_capture_errors - assert ("argument
--operation: invalid choice: 'search' (choose from 'annotate', 'counts', 'in
dex', 'no_input_styles')\n" and "argument --o...nput_styl...
FAILED test/test_wikimedia.py::WikipediaTest::test_wikipedia_page_from_wikid
ata - urllib.error.URLError: <urlopen error [WinError 10060] A connection at
tempt failed because the connected party did not properly respond after a pe
riod ...
====================================== 6 failed, 419 passed, 99 skipped, 1 x
failed, 1 warning in 2155.67s (0:35:55) ======================================
===
```

output obtained when run pytest on amilib

```
(venv) (base) C:\Users\Dell\pygetpapers>pytest
============================================== test session starts ==========================================
=========
platform win32 -- Python 3.12.7, pytest-8.4.1, pluggy-1.6.0
rootdir: C:\Users\Dell\pygetpapers
plugins: anyio-4.9.0
collected 10 items

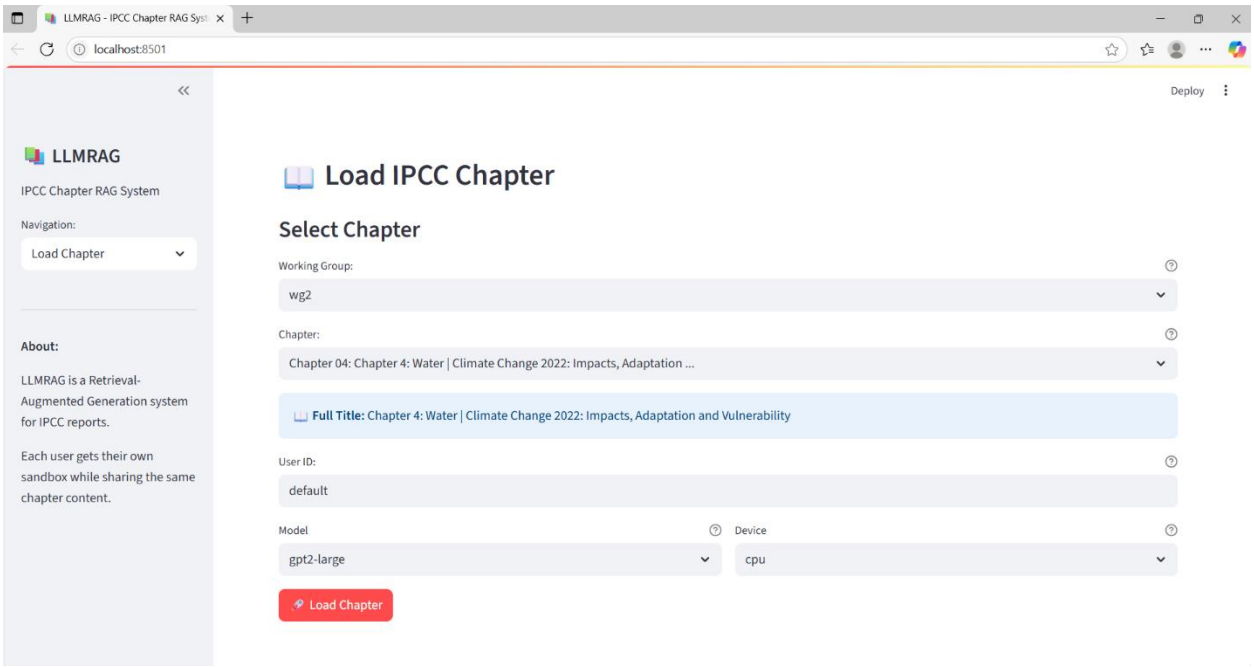
tests\_test.py ...F.FF...
   [100%]
```

output obtained when run pytest on pygetpapers

3.5 Streamlit chatbot debugging and collaboration

Collaborated with team to ensure that the Semantic Climate chatbot interface (built with Streamlit) worked seamlessly on Windows platforms.

- Identified platform-specific issues related to file paths and library compatibility.
- Tested code revisions locally on Windows and suggested fixes (e.g., path handling, dependency adjustments).
- Validated chatbot functionality, ensuring that users on Windows could query climate documents through the Streamlit interface.



Semantic Climate chatbot interface

3.6 Applying Tools to Chapter 4

In this phase of the internship, I explored how emerging tools from the #semanticClimate ecosystem could be applied to IPCC WGII Chapter 4 ("Water") to support advanced document retrieval and interactive knowledge access. This involved using the latest versions of **Pygetpapers:** I worked with the **latest version of Pygetpapers**, a command-line tool designed for retrieving research articles from open repositories such as Europe PMC. The updated version now includes:

- Integration with DataTables: Making the results easier to navigate, filter, and analyze.
- Automatic conversion of XML to HTML: Enhancing human readability and facilitating semantic parsing.
- Improved metadata parsing: For cleaner extraction of abstracts, author details, and publication info.

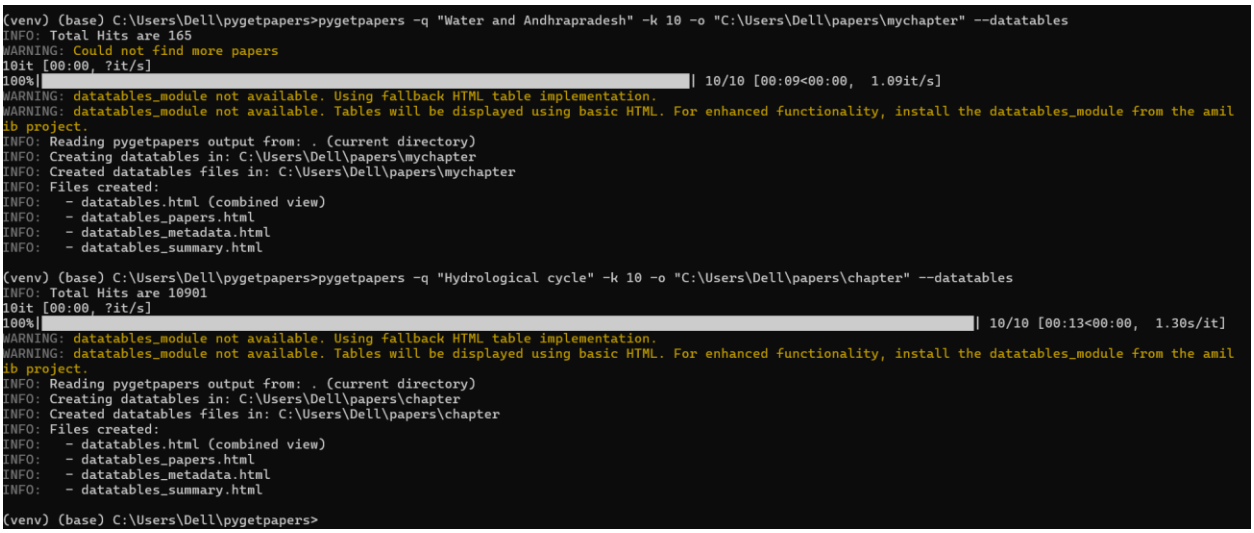
Using **keywords extracted from Chapter 4**, I ran automated queries to collect relevant supplementary literature related to water, climate change, hydrology, and adaptation. This curated set of documents supports both dictionary enrichment and downstream AI analysis tasks.

LLMRAG:

I also contributed to testing and evaluating LLMRAG, a prototype tool that combines language models with retrieval-based methods to answer user queries about climate reports and scientific articles. By applying it to Chapter 4 and, I assessed how effectively it could identify relevant passages, return accurate answers, and support context-aware dialogue.



Setting up pygetpapers using command line



Applying pygetpapers to chapter4

3.7 Contribution to FSCI 2025 Activities

As part of Semantic Climate's involvement in the FORCE11 Scholarly Communication Institute (FSCI), I contributed to participant support and educational resources.

- Prepared general instructions for FSCI participants to engage with Semantic Climate tools.
- Summarized the Cochrane AI RAISE guidance in a simplified PDF format for broader accessibility.
- Created a video presentation summarizing the content and structure of IPCC WGII Chapter 4, intended as an educational asset for newcomers to climate policy and research.

3.8 Documentation of workflow on github

To promote transparency, reproducibility, and collaborative learning, I documented my entire workflow on GitHub, in line with Semantic Climate's commitment to open science and community knowledge sharing.

- Created detailed Markdown guides for setting up the environment, executing keyword extraction and dictionary creation scripts, and interpreting outputs.

- Included code snippets, command-line examples, and troubleshooting notes to ensure that others could replicate and learn from the process.
- Maintained a daily progress log, capturing all tasks completed during the internship, including technical milestones, testing updates, and collaborative contributions.
- Ensured that the documentation was consistent with Semantic Climate's values of openness, accessibility, and transparent communication.

This repository serves as both a technical guide and a personal learning journal, reflecting my end-to-end contributions to the project.

4. Key Deliverables

All code, scripts, visualizations, and documentation produced during the internship are available in the Semantic Climate repository:

4.1 Contents in Github Repository:

Repository Path: semanticClimate/internship_sC (branch: Haarthi)

semanticClimate/internship_sC at Haarthi

The repository includes a well-organized collection of folders and files representing the full scope of the work carried out during the internship. Key contents include:

- Files related to extraction of wordlist in the keyword_extraction folder
- Final wordlist of the chapter which contains more than 150 words (IPCC-Ch04-Wordlist)
- Python code used to extract the keywords(keyword_extraction.py) HTML file used to create the dictionary of the chapter(wg2chap04_dict.html)
- Graphviz pipeline and obtained svg file in the graphviz folder.
- Setup and instructions for the keyword extraction and dictionary creation process are mentioned in the readme file.
- General instructions made for FSCI participants are in the FSCI_instructions.md
- My daily progress throughout the internship is mentioned in the daily_progress.md

4.2 Links to the outputs:

Link to the wordlist extracted:

internship_sC/Keyword_Extraction/IPCC-Ch04-Wordlist at Haarthi · semanticClimate/internship_sC

Link to the dictionary created in HTML:

internship_sC/wg2chap04_dict.html at Haarthi · semanticClimate/internship_sC

Link to knowledge graph created using graphviz:

internship_sC/graphviz at Haarthi · semanticClimate/internship_sC

Link to the readme file containing the keyword extraction and dictionary creation process:

internship_sC/Keyword_Extraction_Process.md at Haarthi · semanticClimate/internship_sC

Link to the markdowns of general instructions and my daily progress:

internship_sC/FSCI_Instructions.md at Haarthi · semanticClimate/internship_sC

internship_sC/Daily_Progress.md at Haarthi · semanticClimate/internship_sC

Link to the video presentation:

https://github.com/semanticClimate/assisted-literature review/blob/main/presentations/video_presentation.md

5.Tools & Technologies

To achieve the project's goals of transforming static IPCC reports into dynamic, structured, and accessible resources, a range of tools and technologies were utilized. These tools supported tasks such as keyword extraction, dictionary generation, document conversion, and testing.

Tools used to accomplish the aforementioned objectives include, but are not limited to:

- ***pygetpapers:*** A valuable text mining tool that interacts with open access scientific repositories, systematically acquiring relevant articles. It comes with the packages pygetpapers and download tools which provide various functions to download, process and save research papers and their metadata. [2].
- ***docanalysis:*** is a command-line tool that processes document collections and performs text analysis. It uses custom code along with Python tools like NLTK, and it can use spaCy or scispaCy for extracting and annotating entities. The tool creates summary data and word lists as output. [3].
- ***amilib:*** amilib has tools for finding, cleaning, converting, searching, republishing legacy documents. It is a Python library designed for document processing,and dictionary creation. We can create dictionaries using amilib from existing set of words. The library simplifies data extraction and manipulation, offering a user-friendly interface for processing data formats like HTML and XML. It ensures that complex operations like term marking and dictionary building can be performed with minimal coding effort.[4].

A non-exhaustive list of software and technologies critical to our functioning:

- Git: A distributed version control system that tracks versions of files.
- GitHub: A developer platform used extensively for hosting our code. It is accessible to the public and easy to collaborate on.
- Python: A powerful high-level programming language that is ideal for scripting and rapid application development.
- VS Code or PyCharm: A highly efficient Integrated Development Environment (IDE) that allows coding, testing and debugging for software development.
- Google Colab: A hosted Jupyter Notebook service that requires no setup to use and provides free of charge access to computing resources, including GPUs and TPUs.
- Slack: A cloud-based team platform for seamless communication and coordination

Importance of semantic toolkit: The semantic toolkit developed by #semanticClimate plays a central role in enhancing the accessibility and utility of climate science reports. IPCC documents are often published in dense, PDF formats that are difficult to search, analyze, or link contextually. The semantic toolkit addresses this by:

- Structuring content semantically (e.g., converting PDF to HTML),
- Extracting high-value keywords and terms,
- Linking those terms to open knowledge bases (such as Wikidata or Wikipedia),
- Generating visual navigational aids like knowledge graphs,
- Creating contextual dictionaries to aid in interpretation.

Ultimately, the semantic toolkit empowers researchers, educators, and the public to interact with climate science in more meaningful and accessible ways[5].

6. Conclusion

The Eight weeks of my internship with #SemanticClimate have been an enriching and insightful journey, blending technical skill-building with meaningful contributions to climate science. I gained hands-on experience with open science tools such as Pygetpapers, amilib and worked on the semantic processing of Chapter 4 from the IPCC Working Group II report.

My contributions included creating a structured keyword wordlist, developing a contextual dictionary, and designing a knowledge graph to support semantic search and accessibility. I also documented the extraction and dictionary-building process in Markdown for GitHub, and applied document retrieval tools to Chapter 4 using Pygetpapers and LLMRAG. Additionally, I participated in alpha testing of core tools, collaborated on interface testing for a chatbot prototype, and created participant-facing resources and a video presentation for FSCI 2025.

This internship has significantly strengthened my technical skills and deepened my understanding of how open, structured scientific knowledge can support climate action. It has provided me with valuable exposure to real-world research workflows, collaborative open-source practices, and the intersection of technology and environmental science.

7. References

[1] IPCC Sixth Assessment Report (WGII – Chapter 4). https://www.ipcc.ch/report/ar6/wg2/

[2] pygetpapers – GitHub Repository. https://github.com/petermr/pygetpapers

[3] docanalysis – GitHub Repository. https://github.com/petermr/docanalysis

[4] amilib – GitHub Repository. https://github.com/petermr/amilib

[5] semanticClimate . https://semanticclimate.github.io/p/en/